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MAST20030: Differential Equations
Tutorial: Week 12

Question 1

(a) Calculate the Fourier transform of $f(x) = \mathrm{e}^{-2|x|}$ using the integral definition (i.e., do not just quote the result from the tables).

(b) Compute the Fourier transform of

$$f(x) = \begin{cases} -5, & -1 \leqslant x < 0 \\ 5, & 0 \leqslant x \leqslant 1 \\ 0, & \text{otherwise} \end{cases} .$$

Question 2

Consider the second order differential equation

$$G''(x) - G(x) = \delta(x) \tag{1}$$

where $\delta(x)$ is the Dirac delta function.

(a) Find the complimentary solution of Eq. (1).

(b) Take the Fourier transform of both sides of Eq. (1) and hence find $\hat{G}$, i.e., the Fourier transform of G.

(c) Take the inverse Fourier transform of your result in (b) to find a particular solution of Eq. (1). The solution you have found is called the Green's function of the linear operator $\dfrac{\mathrm{d}^2}{\mathrm{d}x^2} - 1$.

Question 3

Consider the second order differential equation

$$u''(x) - u(x) = f(x), \tag{2}$$

where $f(x)$ is left unspecified. Using your result from Q2, and the convolution property for Fourier transforms, show that a particular solution to (2) is given by

$$u(x) = \int_{-\infty}^{\infty} G(x - x')f(x')\mathrm{d}x'.$$

This equation shows that the function $G(x, x')$ is able to transport information about the source distribution $f(x')$ to the point in question x.

Question 4

Consider the following IBVP for the 1D heat equation,

$$\dot{u}(x,t) = \alpha u''(x,t) \qquad \text{with} \quad x \in \mathbb{R},\ t > 0\ ,$$

$$u(x,0) = f(x) \qquad \text{for } x \in \mathbb{R}\ ,$$

$$\lim_{x \to \pm\infty} u(x,t) = 0 \qquad \text{for } t > 0\ ,$$

as discussed in the lecture. Show that the (unnormalized) average temperature

$$\overline{u}(t) = \int_{-\infty}^{\infty} u(x,t)\, dx$$

is constant in time and thus amounts to a conserved quantity.